METRIC
DOD-C-63982(AR)
29 October 1982

MILITARY SPECIFICATION

CARTRIDGE, 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789

This specification is approved for use by the U.S. Army Armament Research and Develpment Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and packaging of one type of electric primed cartridges designated Cartridge, 30MM HEDP, M789.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
DOD-C-63980	-	Cartridge, 30MM, Target Practice M788

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-414	-	Samples Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1167	-	Ammunition Data Cards
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKING DRAWINGS

11825954	-	Cartridge, 30MM, HEDP, M789
11825947	-	Cartridge 30MM TP M788
11825969	-	Barrel, Gun 30MM - M230 Gun
10542565	-	Container Shipping & Storage M592
11825879	-	Link, Cartridge 30MM - M29
11825890	-	Container, Shipping and Storage 30MM, Bulk Packed
11825891	-	Container, Shipping and Storage 30MM, Linked, RH-Packed
11825892	-	Container, Shipping & Storage, 30MM Linked, LH Packed
11825893	-	Marking, Shipping and Storage Container

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Code of Federal Regulations

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

3.1.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg. 11825954), all associated drawings, and with all requirements specified in applicable specifications.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the Contracting Officer and no deviation from that process shall be made without prior approval (ref 6.3).

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.4 Projectile extraction. The projectile, when subjected to a bullet pull using a chamber simulating ring at a rate of 25mm per minute ±10 percent, shall have a retention force such that the mean value less three times the standard deviation shall be greater than 8 kilo-Newtons.

3.5 Fuze torque. The projectile shall withstand a torque as specified by Dwg. 11825955.

3.6 Electrical resistance, primed case. The electrical resistance of the primed case shall be as specified on Dwg. 11825901.

3.7 Propellant charge. The propellant charge used in each cartridge shall be maintained to meet the ballistic requirements.

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3.8 Muzzle velocity. The projectile shall have a mean muzzle velocity of 795-815 meters per second (m/s) at 21°C ± 5°C. The standard deviation shall not exceed 15 m/s at temperatures of -54°C ± 5°C, 21°C ± 5°C, and 71°C ± 5°C. Following environmental exposure, the mean muzzle velocity shall not increase or decrease from the mean muzzle/velocity at 21°C ±5°C by more than 30 m/s.

3.9 Case mouth pressure. The mean peak pressure shall not exceed 270-310 MPa at 21°C ±5°C. The mean peak pressure shall not exceed 415 MPa at temperatures of -54°C ± 5°C and 71°C ± 5°C respectively. Following environmental exposure, the mean peak pressure shall not increase or decrease from the mean peak pressure at 21°C ±5°C by more than 60 MPa.

3.10 Environmental conditions. The cartridge shall not be damaged, deteriorated, or exhibit degraded performance beyond the limits of this specification after exposure to the following environmental conditions.

3.10.1 Temperature, humidity and altitude. Temperature/humidity/altitude cycling conditions in accordance with MIL-STD-810, Method 518, with the temperature ranging from -54°C to 65°C, relative humidity up to 95 percent, and pressure conditions equivalent to 15.2 kilometers altitude.

3.10.2 Salt fog. Exposure to salt sea atmosphere specified in MIL-STD-810, Method 509.

3.10.3 Aircraft vibration. Vibration as specified in MIL-STD-810, Method 514 except as modified in 4.5.7.3.

3.10.4 Waterproofness. Immersion for one hour in water under a pressure of 0.1 MPa ± 10% at 21°C.

3.11 Accuracy. The cartridge when fired in a single shot test barrel (Dwg 11825969) shall be such that the standard deviation of the impact points about the mean impact point shall not exceed 1.35 mil.

3.12 Action time. The mean action time plus 3 standard deviations shall not exceed 4.0 milliseconds (ms) at -54°C, 21°C and 71°C. The maximum individual action time shall not exceed 4.0 ms at -54°C ± 5°C, 21°C ± 5°C and 71°C ± 5°C. The action time is defined as the time period between the application of electric current to the primer and the exiting of the projectile from the muzzle.

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3.13 Function and casualty. The cartridge shall function as required and there shall be no separation or breakup of cartridge parts during the firing sequence. In addition when fired in the automatic gun there shall be no gun stoppages caused by the ammunition. See 6.11.

3.14 Penetration.

3.14.1 Dynamic penetration. When fired from a Barrel, Gun 30MM - M230 Gun (Dwg. 11825969) the cartridge shall completely penetrate the target specified in 4.5.14 with the reliability of 70% (complete penetration is evidenced by an exit hole at the rear of the target).

3.14.2 Static penetration. Projectiles shall be tested against 3" (76.2mm) thick rolled homogeneous armor of spin rates of 660, 770 and 980 revolution per second. The mean penetration depth at each spin rate shall exceed the mean value less two standard deviations established during First Article tests.

3.15 Normal impact function. When fired against 1.6MM (nominal thickness) 2024T3 aluminum alloy sheet the cartridge shall function high order.

3.16 Contamination of energetic components. Extreme care shall be exercised to avoid contamination of primers and propellant by oil, grease or other foreign matter since such contamination may degrade performance.

3.17 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.17.1 Process defects. The cartridges shall be free of defects such as folds, wrinkles, buckles, splits, severe scratches, cuts, dents, perforations and other forms of substandard workmanship.

3.17.2 Cleanliness. The cartridge and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article tests.

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE: Cartridge and Components		SHEET 1 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liner - Proj., 30MM, HEDP, M789 (Dwg. 11825960) Examination for defects	32		3.1.2	4.4.2.2
	Retainer, Proj., 30MM, HEDP, M789 (Dwg. 11825961) Examination for defects	32		3.1.2	4.4.2.3
	Pellet, Explosive, Proj., 30MM, HEDP, M789 (Dwg. 11825959) Examination for defects	32		3.1.2	4.4.2.4
	Pellet, Booster, Proj., 30MM, HEDP, M789 (Dwg. 11825962) Examination for defects	32		3.1.2	4.4.2.5
	Charged Body Assembly Proj., 30MM, HEDP, M789 (Dwg. 11825956) Examination for defects	32		3.1.2	4.4.2.6
	Flashtube Assembly (Dwg. 11825903) Examination for defects	32		3.1.2	4.4.2.7
	Case Cartridge Assembly (Dwg. 11825091) Examination for defects	32		3.1.2	4.4.2.8

NOTES:

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TABLE I. First article tests.

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE: Cartridge and Components		SHEET 2 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 30MM, HEDP, M789 (Dwg. 11825955)				
	Examination for defects	32		3.1.2	4.4.2.9
	Cartridge, 30MM, HEDP, M789 (Dwg. 11825954)				
	Examination for defects	32		3.1.2	4.4.2.1
	Major 101-108			3.1.2	4.4.2.1
	Minor 201-207			3.1.2	4.2.2.1
	Test (Sht 3 of 3)	430			

NOTES:

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TABLE I. First article tests.

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Cartridge, 30MM, HEDP, M789				11825954
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
See 1/					
A	Projectile Extraction	20		3.4	4.5.8
A	Propellant Charge	20		3.7	4.5.9
A	Propellant Contamination	20	ACC-0 REJ-1	3.17	4.5.17
B	Velocity 2/	60		3.8	4.5.2
B	Chamber Pressure 2/	60		3.9	4.5.3
B	Action Time 2/	60		3.12	4.5.4
B	Function & Casualty (Mann Barrel)	60	ACC-0 REJ-1	3.13	4.5.5
	Temperature/Humidity/Altitude	20	ACC-0 REJ-1	3.10.1	4.5.7.1
	Salt Spray	20	ACC-0 REJ-1	3.10.2	4.5.7.2
	Aircraft Vibration	20	ACC-0 REJ-1	3.10.3	4.5.7.2
B	Accuracy 2/	60		3.11	4.5.10
C	Dynamic Penetration 3/	25	ACC-4 REJ-5	3.14	4.5.14
C	Normal Impact Function 3/	45	ACC-1 REJ-2	3.15	4.5.15
	Waterproofness	20	ACC-0 REJ-1	3.10.4	4.5.7.4
	Function & Casualty (Auto gun) 4/	200	ACC-0 REJ-1	3.13	4.5.5

NOTES: 1/ Tests identified by the same letter shall be conducted on the same samples. 2/ 40 cartridges at 21°C and 10 cartridges each at -54°C & 71°C. Data collected at -54°C and 71°C will be recorded for information only. 3/ Fired at 21°C. 4/ Gun specified in the contract or purchase order.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Cartridge cases from one lot interfix number from one manufacturer.

b. Projectile metal parts from one lot interfix lot number from one manufacturer.

c. Primers from one lot interfix lot number from one manufacturer.

d. Propellant from one batch lot number from one manufacturer and which has been ballistically tested in accordance with 4.5.9 to establish the propellant charge.

e. Fuzes from one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. MIL-STD-1235 may be used as an alternative only upon approval of the contracting officer.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, HEDP M789 1/	1 OF 3	11825954
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
101	Case to projectile crimp missing	2/	100%	3.1.2	Visual
102	Split or perforated case	2/	100%	3.1.2	Visual
103	Exterior profile (max) 3/	2/	100%	3.1.2	Gage
104	Overall length (max)	2/	100%	3.1.2	Gage
105	Case crimp diameter (max) 4/	2/	100%	3.1.2	Gage
106	Max gap case-projectile interface 6/	See 4.4.2a	.25%	3.1.2	Gage
107	Case projectile diameter between crimps	4.4.2a	.25%	3.1.2	Gage/Visual
108	Fuze nose cap damaged		.25%	3.1.2	Visual
	Tests				
109	Projectile extraction *	Table II	ACC-0 REJ-1	3.4	4.5.8
110	Presence of waterproof seal * (projectile to case)			3.1.2	Visual
111	Propellant contamination *			3.17	4.5.17
112	Propellant charge & propellant weight * 5/	2/	100%	3.7/3.16	4.5.9
	Ballistic performance (21°C)				
113	Velocity **	Table III		3.8	4.5.2

NOTES:

* Test may be conducted using the same sample cartridges.

** Test may be conducted using the same sample cartridges.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, HEDP M789 1/	2 OF 3	11825954
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
		Table III			
114	Chamber pressure **		100%	3.9	4.5.3
115	Action time **		100%	3.12	4.5.4
116	Accuracy **	25	100%	3.11	4.5.10
117	Waterproofness	20	100%	3.10.4	4.5.7.4
118	Dynamic penetration	75	100%	3.14.2	4.5.14.2
119	Normal impact function	45	ACC-1 REJ-2	3.15	4.5.15
120	Function & Casualty	Table III	Table IV		
	Ballistic performance (-54°C & 71°C)				
121	Velocity **	Table IV		3.8	4.5.2
122	Chamber pressure **			3.9	4.5.3
123	Action time **			3.12	4.5.4
124	Function & Casualty (Mann Barrel)	Table IV	Table VI	3.13	4.5.5
Minor					
201	Protective coating damaged or incomplete 7/	4.4.2a	.65%	3.1.2	Visual
202	Incorrect type cartridge	4.4.2a	.65%	3.1.2	Visual
203	Foreign material (except corrosion)	4.4.2a	.65%	3.1.2	Visual
204	Workmanship	4.4.2a	.65%	3.16	Visual

NOTES:

** Test may be conducted using the same sample cartridges.

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CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 11825954
4.4.2.1	Cartridge, 30MM, HEDP M789 1/				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
205	Folds, wrinkles, buckles, deep scratches, cuts and dents	4.4.2a	.65%	3.1.2	Visual
206	Corrosion	4.4.2a	.65%	3.1.2	Visual
207	Marking incorrect, illegible or missing	4.4.2a	.65%	3.1.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Liner, Proj., 30MM, HEDP, M789	1 OF 2	11825960
			NEXT HIGHER ASSEMBLY: 11825956

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.4.2			
Major:					
101	Diameter, inside at surface "X"		.25%	3.1.2	Gage
102	Diameter, outside at surface "X"		.25%	3.1.2	Gage
103	Large outside diameter, max.		.25%	3.1.2	Gage
104	Large inside diameter, min.		.25%	3.1.2	Gage
105	Concentricity of outside diameter small ends		.25%	3.1.2	Gage
106	Location of large datum diameter		.25%	3.1.2	Gage
107	Location of small datum diameter		.25%	3.1.2	Gage
108	Thickness T, section A-A & B-B		.25%	3.1.2	Gage
Minor:					
201	Overall length	5	ACC-0 REJ-1	3.1.1	Gage
202	Large outside diameter, min	5	ACC-0 REJ-1	3.1.1	Gage
203	Large inside diameter, max	5	ACC-0 REJ-1	3.1.1	Gage
204	Radius, A, section A-A & B-B	5	ACC-0 REJ-1	3.1.2	Gage
205	Depth C, section A-A & B-B	5	ACC-0 REJ-1	3.1.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 11825960
4.4.2.2	Liner, Proj., 30MM, HEDP, M789				NEXT HIGHER ASSEMBLY 11825956
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
		See 4.4.2a			
206	Depth, D, section A-A & B-B		.40%	3.1.2	Gage
207	Angle E, section A-A & B-B		.40%	3.1.2	Gage
208	Angle F, section A-A & B-B		.40%	3.1.2	Gage
209	Workmanship		.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Retainer, Proj., 30MM, HEDP, M789	1 OF 1	11825961
			NEXT HIGHER ASSEMBLY 11825956

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined	See 4.4.2a			
Major:					
101	Outside diameter		.25%	3.1.2	Gage
102	Height of retainer		.25%	3.1.2	Gage
103	Thread diameter		.25%	3.1.2	Gage
Minor:					
201	Length of thread		.65%	3.1.2	Gage
202	Evidence of poor workmanship		.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825959
4.4.2.4	Pellet, Explosive, Proj., 30MM, HEDP, M789				NEXT HIGHER ASSEMBLY 11825956
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined	See 4.4.2a			
Major:					
101	Weight		.25%	3.1.2	Scale/Balance
Minor:					
201	Outside diameter		.65%	3.1.2	Gage
202	Length		.65%	3.1.2	SMTE 1/

NOTES:

1/ Standard measurement test equipment.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Pellet, Booster, Proj., 30MM, HEDP, M789	1 OF 1	11825962
			NEXT HIGHER ASSEMBLY 11825956

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined	See 4.4.2a			
Major:					
101	Weight		.25%	3.1.2	Scale/Balance
Minor:					
201	Outside diameter		.65%	3.1.2	Gage
202	Length		.65%	3.1.2	SMTE

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825956
4.4.2.6	Charged Body Assembly Proj., 30MM, HEDP, M789				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Min depth and presence of retainer	1/	100%	3.1.2	Gage
102	Presence of seal	1/	100%	3.1.2	Visual
103	Explosive of threads at indicated length	1/	100%	3.1.2	Visual
104	Weight	1/	100%	3.1.1	Gage
Minor;					
201	Workmanship	See 4.4.2a	.65%	3.16	Visual

1/ Inspection verification of prior 100% inspections shall be performed per (a) by utilization of Government approved machine masters or (b) MIL-STD-105, AQL = 0.25, to determine sample size, minimum sample size - 315, zero defects allowed.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825903
4.4.2.7	Flashtube Assembly				NEXT HIGHER ASSEMBLY 11825901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Ignitor material missing or incorrect (Broken pellets permissible)	See 4.4.2a	.25%	3.1.2	Visual
Minor:					
201	Evidence of poor workmanship		.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825901
4.4.2.8	Case Cartridge Assembly				NEXT HIGHER ASSEMBLY 11825954
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major:					
101	Depth of primer seating	2/	100%	3.1.2	Gage
102	Primer/Case Resistance	2/	100%	3.1.2	Gage
103	Presence of Igniter Tape Seal	2/	100%	3.1.2	Gage
104	Dimpling of Cartridge Case (see note)	2/	100%	3.1.2	Visual
105	Loose Primer	4.4.2a	.25%	3.1.2	Manual
106	Corrosion	4.4.2a	.25%	3.1.2	Visual
Minor:					
201	Primer Waterproofing Missing or Incomplete	4.4.2a	65%	3.1.2	Visual
202	Incomplete or Missing Primer Crimp	4.4.2a	65%	3.1.2	Visual
203	Evidence of Poor Workmanship	4.4.2a	65%	3.16	Visual

NOTES:

Note: Inspection shall be performed at cartridge level.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Projectile, 30MM, HEDP, M789	1 OF 1	11825955
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:		See 4.4.2a			
101	Disassembly torque test of fuze		.25%	3.1.2/3.5	4.5.20
102	Gap between fuze and body, max.		.25%	3.5	Gage
103	Bourrellet diameter, max.		.25%	3.1.2	Gage
Minor:					
201	Workmanship		.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825890 or 1182589 ~~or 1182589~~ NEXT HIGHER ASSEMBLY
4.4.2.10	Container, Shipping and Storage, 30MM Packed				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1	Link improper (anyone of the four (4) tabs of link not in groove, link inverted; link damaged; link distorted)	2 boxes/ lot	100%	3.1.2	Visual
Major:					
101	Cartridge inverted		100%	3.1.2	Visual
102	Cushions, fillers, liners and separators missing		100%	3.1.2	Visual
103	Tubes missing (if applicable)		100%	3.1.2	Visual
104	Link orientation (LH-RH) (if applicable)		100%	3.1.2	Visual
105	Contents loose		100%	3.1.2	Visual
Minor:					
201	Metallic seal missing		100%	3.1.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63982(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Marking on Shipping and Storage Container M592, 30MM Cartridge	1 OF 1	11825893
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:	None defined				
Minor:					
201	Marking misleading or unidentifiable		0.65%		Visual

NOTES:

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NOTES FOR PARAGRAPH 4.4.2.1 & 4.4.2.8

1/ Visual standards/machine masters established and approved by the Government as defined in 6.4, shall be used in evaluating cartridge case and cartridge visual defects. Major components shall have previously been accepted in accordance with the acceptance criteria defined in the specification listed in the applicable drawing.

2/ Inspection verification of prior 100% inspections shall be performed per (a) by utilization of Government approved machine masters or (b) MIL-STD-105, AQL = 0.25, to determine sample size, minimum sample size = 315, zero defects allowed.

3/ A chamber gage in accordance with Dwg. 9354298 will be used. The maximum force to insert the cartridge into the gage shall comply with either of the following criteria.

a. Apply a static force of not to exceed 270 Newtons.

b. Apply a dynamic force by hand (no mechanical aid) which does not damage the cartridge.

4/ The lot shall not be penalized during inspection verification for oversize conditions provided that:

a. The crimp is visually complete for 360° and,

b. A sample quantity of the discrepant cartridges in accordance with Table II meet the projectile extraction requirement of 3.4

5/ A cartridge weighing less than the value determined by the formula specified in 3.16 shall be rejected. The 100% inspection method may be by verification of weight of each cartridge or by probe for a fixed minimum propellant height in a charged cartridge case. If the probe method is used, the following method shall be used to establish the minimum height for each cartridge lot:

a. Establish the mean charge weight in accordance with 4.5.9.

b. Prior to any loading charge a minimum of 25 cases to the specified weight, measure each case and calculate the mean propellant height for the propellant batch to be loaded.

c. Repeat step b with a propellant charge weight 1 gram less less than the established mean charge weight.

d. Adjust the probe height such that any cartridge in which the propellant charge is 1 gram below the mean weight will be rejected.

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e. Verify the probe height at the beginning of of production and every 4 hours of production or portion thereof, and at the end of production of the lot. Any probe light failure shall cause rejection of the cartridges loaded since the last acceptable probe height verification.

f. Verify the proper propellant height of the end of lot production by repeating steps b and c and comparing the established probe height.

6/ A gap of 0.25mm or less at any point on the circumference is acceptable.

7/ The lot shall not be penalized for anodize crazing in the cartridge case crimp grooves (see 6.12).

TABLE II. Sample size for projectile extraction test.

Lot Size	Sample Size
0 - 800	10
801 - 1,300	15
1,301 - 3,200	20
3,201 - 8,000	25
8,001 - 22,000	30
22,001 and over	35

Reference: MIL-STD-414, Level 2, Table A-2
Except for lots of less than 800 units.

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TABLE III. Sample size for 21°C ballistic tests.

Lot Size	First Sample Size	Second Sample Size
0 - 800	30	35
801 - 1,300	35	40
1,301 - 3,200	40	50
3,201 - 8,000	50	75
8,001 - 22,000	75	100
22,001 - 110,000	100	140
110,000 - 550,000	150	190
Greater than 550,000	200	255

Reference MIL-STD-414, Level 4, Table A-2
Except for lots of less than 800 units.

TABLE IV. Sample size for 71°C and -54°C ballistic tests.

	Sample Size for Each Temperature	
Delivered Lot Size	First Sample Size	Second Sample Size
0 - 800	10	10
801 - 1,300	10	10
1,301 - 3,200	10	15
3,201 - 8,000	15	20
8,001 - 22,000	20	20
Greater than 22,000	25	25

Reference: MIL-STD-414, Level 1, Table A-2
Except for lots of less than 3200 units.

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TABLE V. Firing defects 1/

Item	Defect	AQL	Classification
Cartridge	Misfire due to cartridge 2/	0.065	Major
	Failure to chamber due to cartridge 3/	0.065	Major
	Failure to extract due to cartridge 4/	0.065	Major
	Projectile remaining in bore 5/	0.065	Major
	Early Burst 6/	0.0065	Critical
Primer	Blown primer 7/	0.25	Major
	Loose primer 8/	0.65	Minor
	Primer leak 9/	0.65	Minor
Cartridge Case	Split or rupture in the cartridge case 10/	0.65	Major
	Detached metal from the cartridge case 11/	0.25	Major
	Erosion, internal or external 12/	0.40	Minor
	Burn-through, base 13/	0.065	Major
	Burn-through, major 13/	0.40	Minor
	Burn-through, minor 13/	0.65	Minor
Projectile	Metal parts separation 14/	0.065	Major

1/ These defects apply only to cartridges fired in the approved test weapon.

2/ Misfire. If a misfire occurs, a second firing voltage application is not allowed unless a non-ammunition cause for the misfire is confirmed and corrected.

3/ Failure to chamber. The cartridge cannot be inserted into chamber sufficient distance to allow the breach to be engaged or cartridge cannot be bottomed by the breach within the limits of an axial load of 270 Newtons.

4/ Failure to extract. After firing, the cartridge case cannot be removed without failure of the extractor ring. Foreign debris at cartridge/chamber interface causing cartridge case removal difficulty does not constitute a valid failure.

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5/ Projectile remaining in bore. This defect is defined as a failure of the projectile to exit the barrel after the primer has fired.

6/ Early burst. An early burst is considered to have occurred if the projectile functions prior to impact.

7/ Blown primer. The cartridge case shall be visually examined. A blown primer is a primer which when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the primer pocket are enlarged and deformed.

8/ Loose primer. The cartridge case shall be visually examined. A loose primer is defined as a looseness which permits the fired primer to fall from the primer pocket after the cartridge has been fired, without distortion of the primer pocket.

9/ Primer leaks. Primer leads shall be defined as follows:

a. Suspect. A suspect primer leak is defined as evidence of smudging or discoloration around any part of the periphery of the primer cup, button, insulator, or primer pocket. The fired primer must be removed from the case and disassembled to determine whether actual gas leakage has occurred.

b. Actual. A primer leak suspect, which upon disassembly, shows erosion of the primer insulator, or which shows a definite leakage path such as a crack or split.

c. A dislodged primer button is not cause for rejection.

10/ Split or ruptured cartridge case. The cartridge case shall be visually examined. A split or rupture is acceptable providing proper obturation is obtained.

11/ Detached metal. The cartridge case shall be visually examined. Metal sheared or missing from the exterior profile of a fired cartridge case, such as a rim or neck shear or ignition system component, shall be classed as detached metal. Shavings or metal from the interior wall of the neck in the crimp area shall not be considered as detached metal.

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12/ Mouth erosion. Cartridge case mouth erosion is defined as erosion of an area greater than 160 square millimeters to a depth such that the remaining thickness of the cartridge case mouth is reduced to less than 0.25mm.

13/ Burn-through.

Major. Major burn-through is defined as a hole through the case due to melting which exceeds 2.5mm in any dimension.

Minor. Minor burn-through is defined as a hole through the case due to melting which is 2.5mm or less in any dimension.

Base. Base burn-through is defined as a hole in the base area not to include primer leaks. The base area is defined as the area of the case not supported by the chamber.

14/ Loss of projectile structual integrity (metal parts only). See 4.5.6.

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TABLE VI. Function and casualty firing defect accept/reject criteria

Lot Size	AQL			
	0.065	0.25	0.40	0.65
501 - 800	0/1 *	0/1 *	0/1 *	0/1 *
801 - 1,300	0/1 *	0/1 *	0/1 *	0/2 1/2
1,301 - 3,200	0/1 *	0/1 *	0/2 1/2	0/2 2/3
3,201 - 8,000	0/1 *	0/1 *	0/2 2/3	0/3 4/5
8,001 - 22,000	0/1 *	0/2 2/3	0/3 3/4	1/3 4/5
22,001 - 110,000	0/1 *	0/2 2/3	0/3 4/5	1/4 6/7
110,001 - 550,000	0/1 *	0/3 3/4	1/4 4/5	2/5 6/7
Over 550,000	0/1 *	1/3 3/4	2/4 6/7	3/6 8/9

* Indicates single sampling plan, i.e., no retest

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4.4.3 Testing.

4.4.3.1 Unlisted firing defects. The lot should be suspended and referred to the contracting officer for disposition if any malfunction or casualty not covered by this specification occurs in any firing test.

4.4.3.2 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.14 herein.

4.5 Test methods and procedures.

4.5.1 Test barrel. Test samples for all firing tests shall be fired from a test barrel conforming to Drawing 11825969. (See 6.6 for criteria for barrel replacement.)

4.5.2 Muzzle velocity. Velocity shall be measured by the use of velocity coils or light screens. The test cartridges shall be conditioned at -54°C, 21°C and 71°C respectively before firing. The mean velocity and standard deviations from the mean of the sample cartridges in each group, shall conform to 3.8. The velocity shall be corrected to the muzzle velocity as follows:

$$V_m = \frac{d_1}{t} + c\left[d_2 + 0.5d_1\right]$$

where:

t = time between screens, seconds

d_1 = distance between screens, meters

d_2 = distance from muzzle to first screen, meters

c = constant of proportionality, 0.75 meters/second/meter

4.5.3 Case mouth pressure. The test rounds shall be conditioned at -54°C, 21°C and 71°C temperatures respectively and fired from a test fixture barrel. Pressure measurements shall be made at the case mouth with the piezo-electric transducer. A permanent record shall be made of the resulting pressure. Data at -54°C and 71°C will be recorded for information only. This record may be a pressure trace, paper print out of the data or other similar instrument reading. The pressure shall conform to the requirements of 3.9.

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4.5.4 Action time. The test rounds shall be conditioned at -54°C, 21°C and 71°C and fired from a test fixture barrel. The time between application of voltage to primer and exit of the projectile from the muzzle shall be measured. The time shall comply with the requirement of 3.12. Any corrected action time of 2.40 milliseconds or less shall be disregarded as an instrumentation error.

4.5.5 Function and casualty. These tests shall be conducted, using the test methods prescribed in AMCR-715-505, Volume 8, Chapter 1, Section 3 as a guide. Witness panels for structural integrity shall be placed approximately 5 meters from the muzzle. A failure is characterized by observance of any of the firing defects listed in Table V. Test rounds, when required, shall be fired in the M230 Gun. Test results shall conform to the requirements of 3.12.

4.5.6 Projectile structural integrity. Projectile structural integrity shall be verified in conjunction with the ballistic tests of 4.5.2, 4.5.3 and 4.5.4. A witness board/card shall be located approximately 5 meters from the gun muzzle. Loss of projectile structural integrity is defined as projectile fracture, break-up, rotating band failure as defined in 6.10, or loss of parts as evidenced by microflash photography and/or hole(s) in the witness board or recovery of fragments.

4.5.7 Environmental tests. Unless otherwise specified, performance after environmental exposure, as specified in the applicable test procedure, shall consist of conducting the tests of 4.5.2, 4.5.3 and 4.5.4, except all test rounds shall be conditioned at 21°C prior to firing.

4.5.7.1 Temperature, humidity and altitude. The cartridge shall be subjected to the temperature/humidity/altitude test of MIL-STD-810, Method 518. After exposure, the test item shall be examined for deterioration and subjected to the tests referenced in 4.5.7.2, 4.5.7.3 and 4.5.7.4.

4.5.7.2 Salt fog. The cartridge shall be subjected to the salt fog test of MIL-STD-810, Method 509 for a period of 48 hours. After exposure, the test item shall be subjected to the tests referenced in 4.5.7.3 and 4.5.7.4.

4.5.7.3 Aircraft vibration. The ammunition shall be subjected to a broad band random vibration profile in accordance with the requirements of MIL-STD-810, Method 514.2, Procedure I, Category C. The test will be conducted with an AAH magazine. The vibration spectrum shall be as shown in Figures 1 and 2. The test item shall be mounted to a test fixture simulating a feed system in a normal airplane mounting. The test duration shall be 45 minutes in each of three mutually perpendicular axis at each of two temperature conditions, -40° ±2.8°C and 65.5 ±2.8°C. Following exposure, the test items shall be subjected to the tests referenced in 4.5.7.4.

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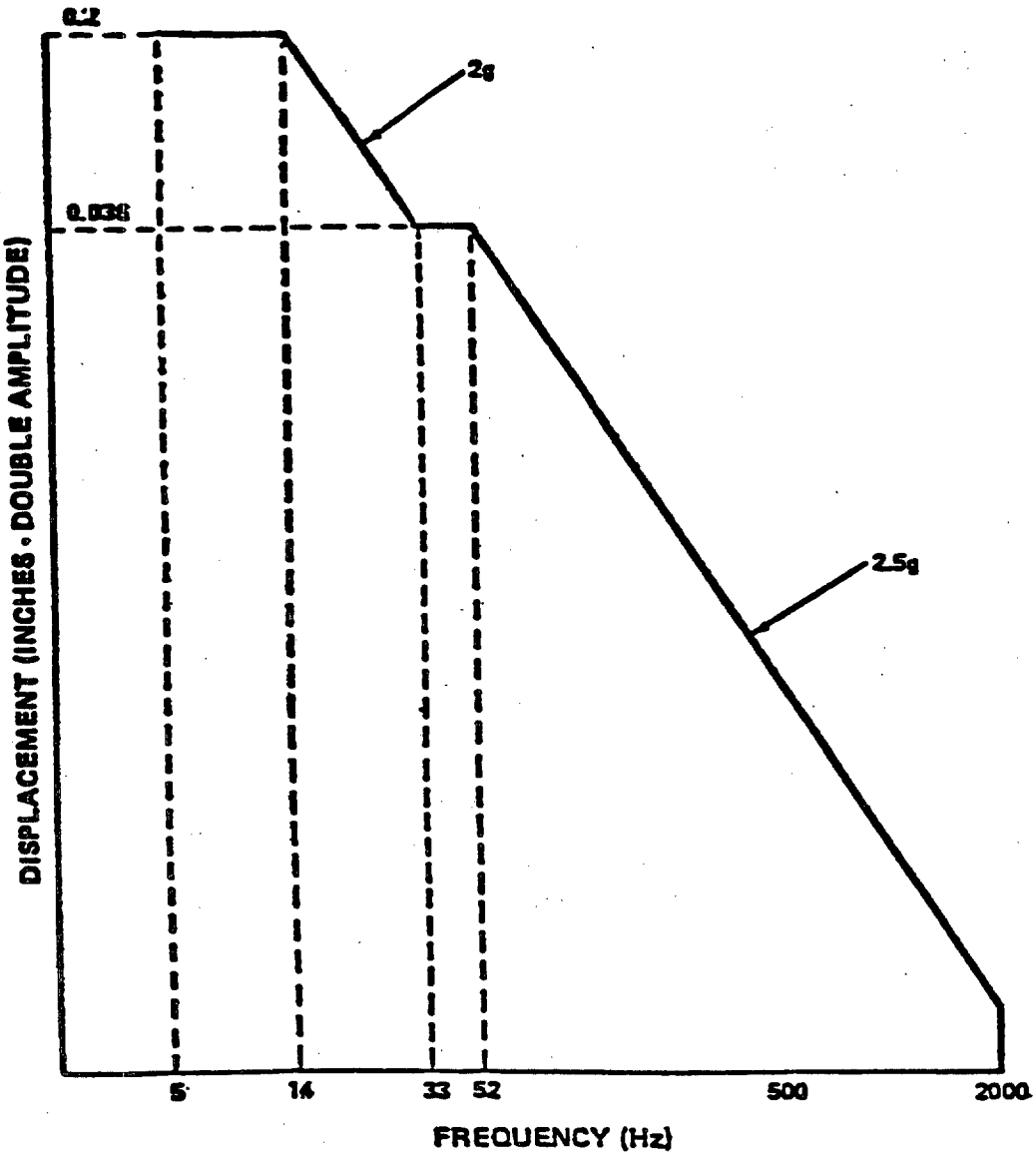


FIGURE 1. Vibration Test Curves for Equipment Installed in Helicopters, Equipment Category C. (Ref. Figure 514.2-3 of MIL-STD-810C)

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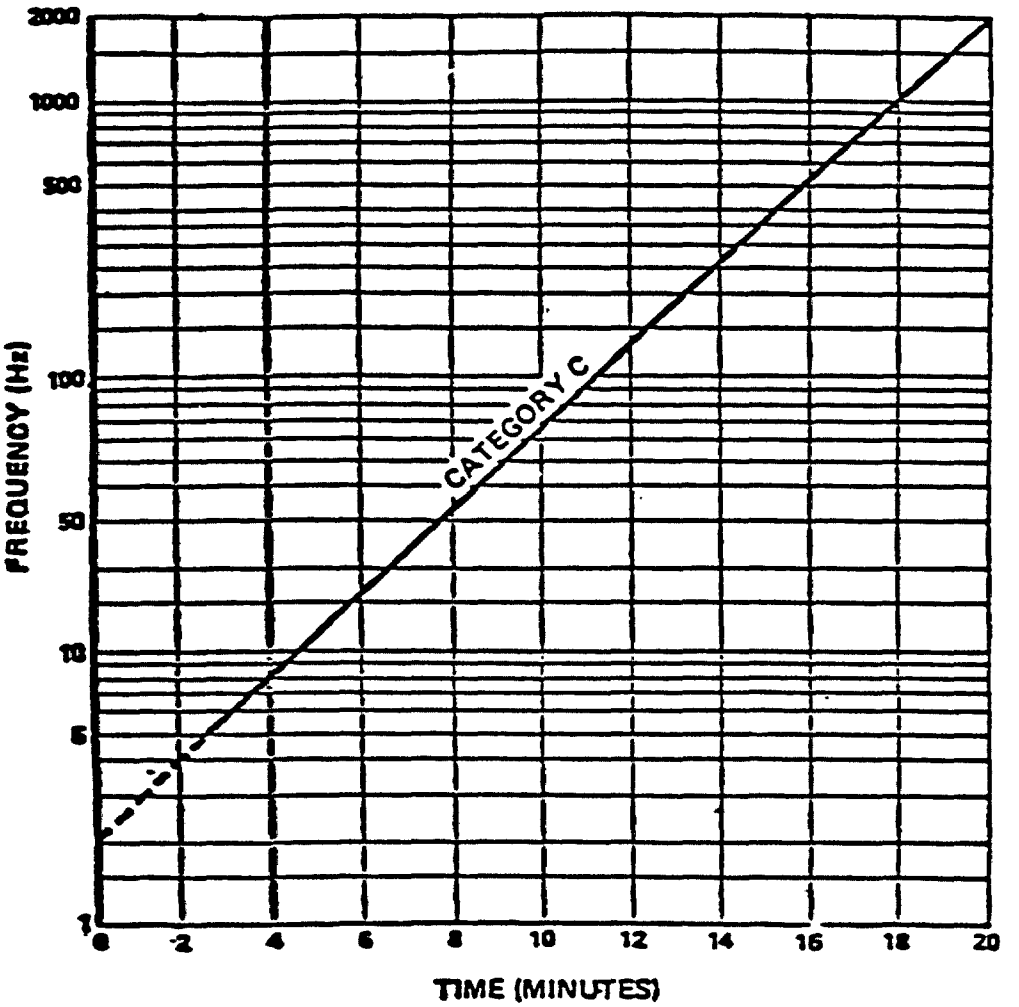


FIGURE 2. Logarithmic Sweep.
(Ref. Figure 514.2-10
of MIL-STD-810C)

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4.5.7.4 Waterproofness. Sample cartridges shall be immersed for one hour in water at the temperature and gauge pressure specified in 3.10.4. The cartridges shall be taken out and wiped dry. The test quantity shall then be tested in accordance with 4.5.7 to verify conformance with the requirements of 3.8, 3.9 and 3.12.

4.5.8 Projectile extraction. The base of the cartridge case shall be secured in a holding fixture and a ring simulating the gun chamber shall be installed in the case neck area. A clamp shall be applied to the projectile so that a tensile load may be applied along the longitudinal axis of the round. An axial load shall be applied at a constant strain rate of 25MM per minute ±10% until the projectile is withdrawn from the cartridge case. The peak axial load shall be recorded and compared to the requirements of 3.4. A case neck separation during this test shall be cause for rejection. A split of greater than 5MM observed during projectile extraction shall be cause for rejection.

4.5.9 Propellant charge. The charge of propellant from a given lot, which is required to meet the performance requirements of 3.8, 3.9 and 3.11 shall be determined by ballistic tests. The minimum charge weight assigned shall be used in determining the minimum cartridge weight in accordance with 3.16.

4.5.10 Accuracy. The test shall be conducted using a single shot test barrel with cartridge conditoned to 21°C. The target shall be positioned at approximately 25 meters from the muzzle. After three barrel fouling shots have been fired the test shots shall be fired without moving the gun or target. The target impacts shall be measured for dispersion from the calculated center of impact as described in AMCR 715-505, Vol 8, Chapter I, Section 4.4. The impact dispersion of these shots wll comply with the requirements of 3.11. Testing shall not be conducted if wind velocity including gusts, is 10 knots or more.

4.5.11 Reference rounds. Reference rounds shall be fired in conjunction with ballistic and accuracy tests. Unless otherwise specified, the reference rounds shall be assembled in accordance with the requirements of Drawing 11825947 and inspected in accordance with this specification DOD-C-63980 with the charge weight controlled to ± 0.2 grams.

4.5.12 Ballistic reference rounds. A minimum of 10 reference rounds shall be fired during each ballistic firing test and the mean muzzle velocity, case mouth pressure and action time recorded. Any individual round readings which vary from the mean by greater than 12 meters per second, 20 MPa or 0.3 milliseconds respectively, shall be disregarded and a new mean velue shall be computed. If more than three rounds are disregarded, additional

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reference rounds shall be fired, provided they are fired during the same day as the applicable tests are conducted. The differences between the mean reference round test values and the reference acceptance baseline values shall not exceed 10 meters per second, 23 MPa, or 0.35 millisecond. Reference round data shall be utilized as defined in 4.5.22.

4.5.13 Accuracy reference rounds. The accuracy reference baseline shall be established during lot acceptance testing of the reference lots in accordance with the tests of Table III, except a minimum of 60 rounds, divided into 20 round groups, shall be used to establish the baseline. The accuracy of each 20 round group shall be determined and an average calculated for the baseline. Reference rounds shall be utilized as defined in 6.19.

4.5.14 Penetration test.

4.5.14.1 Dynamic penetration test. The target shall be 25.4MM nominal thickness rolled homogeneous armor plate. The plate shall be positioned 500 meters $\pm$ 10 meters from the muzzle at an angle of obliquity equal to $50^{\circ} \pm 2^{\circ}$ from the nominal.

4.5.14.2 Static penetration test. Test procedure shall be submitted by the contractor to the PCO for approval. Test procedure shall call for detonating 25 projectiles at each spin rate against a 3" thick RHA target at a stand-off distance which simulates dynamic impact. Failure to meet the requirements of 3.14.2 shall result in referring the lot. Disposition will then be based on performing the test of 4.5.14.1 to insure compliance with 3.14.1.

4.5.15 Normal impact function. The target shall be placed 200 meters $\pm$ 20 meter from the muzzle. Failure of any cartridges to function high order detonation shall be cause for rejection of the lot.

4.5.16 Disassembly torque test of fuze on projectile. After four (4) hours curing time, the projectile shall be placed in a fixture and the Newton/Meter (inch pounds) requirement specified on the applicable drawing shall be applied. Observation shall be made of subject requirement. This test is a non-destructive test.

4.5.17 Propellant contamination. After completion of test specified in 4.5.8, each of the cartridges test will be examined visually for contamination as described in AMCR 715-505, Chapter 1, Section 10, Volume 8.

4.5.18 Retest procedures. Failure of the materiel sample to comply with the velocity pressure and action time requirements shall result in a retest for the failed requirements. Sample size shall be as indicated in the applicable table.

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4.5.19 Accuracy retest criteria. If the Mann barrel dispersion standard deviation σ (test) exceeds the requirements specified in 3.2.1.5 the following procedure shall be followed:

a. Fire a 20 round accuracy test utilizing reference TP ammunition for which the Mann barrel dispersion baseline is known. Calculate the radial standard deviation of the TP rounds ($\sigma_{TP_{ref}}$).

b. Calculate the difference between the TP reference baseline and a 20 round test group.

$$\sigma_{TP_{ref}} - \sigma_{TP_{baseline}} = \Delta_{TP}$$

c. Correct to standard deviation of the test ammunition as follows:

NOTE: $\sigma_{corrected} = \sigma_{test} - \Delta_{TP}$

4.5.20 Abnormal test results (velocity pressure and action time). In the event of an individual abnormal test result, where one of the three critical parameters of muzzle velocity, chamber pressure, or action time was not obtained or is inconsistent with the other two and falls on or outside of the mean plus or minus four sigma value, (calculated without the individual abnormal test result), the abnormal test result may be eliminated from the lot acceptance calculation for the tests of Table II. Calculation of the mean and standard deviation values shall be made both with and without the abnormal data and the specific abnormal data shall be noted. The criteria for acceptance of the lot shall be based on the data base without the abnormal data. If more than three rounds are disregarded additional rounds shall be fired, provided they are fired during the same day and under the same environment as the abnormal test result rounds.

4.5.21 Temperature tolerences. Unless otherwise specified all temperatures are tolerenced $\pm$ 5°C.

4.5.22 Calibration of instrumentation. A reference lot of 30MM TP ammunition will be manufactured. It will be tested and established as a control baseline. The reference lot accuracy will be established by averaging the first three-20 round accuracy tests. Prior to acceptance testing of each lot, a minimum of ten rounds per day will be fired to establish pressure, velocity and action time. Ballistic variation between reference and acceptance tests lots will be adjusted as follows:

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$X_a + (X_{ra} - X_r) =$ corrected X_a

Where X_a is mean of acceptance lot

X_r is mean of reference lot.

X_{ra} is mean of acceptance baseline.

4.5.23 Propellant contamination. After completion of test specified in 4.5.11, each of the cartridges tested will be examined visually for contamination as described in AMCR 715-505, Chapter 1, Section 10, Volume 8.

5. PACKAGING

5.1 Packaging and packing. The cartridge shall be preserved, packaged and packed in accordance with the applicable requirements for the shipment of hazardous items as defined in Title 49, Code of Federal Regulations, Parts 100 through 199 and in accordance with the applicable drawings.

5.1.1 Packing.

5.1.1.1 Linked packing. The linked pack shall consist of linked cartridges packed in the M592 shipping and storage container, (dwg. 10542565) and may include provisions for interconnecting ammunition belts. The links shall conform to Drawing 11825879. The pack shall also include provisions for a right and left hand pack. The packing ratio of left to right belts shall be specified in the contract or purchase order.

5.1.1.2 Bulk packing. The bulk pack shall consist of cartridges packed in the M592 shipping and storage container.

5.1.2 Ammunition container interior packing.

5.1.2.1 Linked packing. The interior packing shall be in accordance with Drawing 11825891 or 11825892 as applicable for right hand or left hand pack.

5.1.2.2 Bulk packing. The interior packing shall be in accordance with Drawing 11825890.

5.2 Marking. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with Dwg 11825893. In addition, exterior containers shall show the caution markings required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition.

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5.3 Documentation with shipment. Each shipment of ammunition shall contain data cards in accordance with MIL-STD-1167, lot number assignment in accordance with MIL-STD-1168, date of manufacture and complete definition of the ammunition including such items as case, primer, flashtube assembly, propellant, charge weights, projectile and lot identification.

5.4 Shipping. When shipments of more than one lot are made at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. This specification covers assembling and packing for Cartridge, 30MM, M789 for firing in the M230 Gun, High Explosive Dual Purpose M789, the Aden MX IV and DEFA series of guns.

6.2 Ordering data. See MIL-A-48078.

6.3 Process deviation. A process deviation is defined as a change in the approved method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Visual defects standards. Components and assemblies which have been rejected during initial production shall be segregated and visual standards established jointly by representatives of the Government and the contractor. Subsequent application of these standards for acceptance inspection shall be specified in the contract or purchase order.

6.5 Accuracy measurement procedure.

a. Select a reference point on the target and establish X and Y Axis through this point. Measure X and Y distance to each impact point.

b. Calculate mean impact point $(\bar{X}, \bar{Y})$, (i.e., $\bar{X} = \Sigma X_i/N, \bar{Y} = \Sigma Y_i/N$).

c. Calculate standard deviations for X and Y.

$$\sigma_x(m) = \sqrt{\frac{\Sigma (X_i - \bar{X})^2}{N-1}} \quad , \sigma_y(m) = \sqrt{\frac{\Sigma (Y_i - \bar{Y})^2}{N-1}}$$

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d. Calculate radial standard deviation (σ_R) based on σ_x and σ_y

$$\sigma_{R(m)} = \frac{\sigma_x + \sigma_y}{2}$$

An alternate method of measuring accuracy is as follows:

$$RSD = \sqrt{(\sigma_x)^2 + (\sigma_y)^2}$$

NOTE: If this alternate method is used, all reference values for accuracy in this specification must be multiplied by 1.5.

e. Adjust calculation for 1000 m target.

$$\sigma_{R(A\ 1000\ m)} = \frac{1000}{(\text{range to target})}\ \sigma_{R(m)}$$

f. convert σ (at 1000 m) to mils dispersion)

1 mil = 1 m at 1000 m

g. Data to be provided shall be individual shot x and y impact location data and calculated dispersion standard deviations σ_x, σ_y, and σ_r.

6.6 Mann barrel. The barrel used for preproduction and lot acceptance testing shall not be used for testing if any of the following conditions exist.

a. The barrel does not meet drawing requirements.

b. Degradation of reference lot muzzle velocity by more than 15 meters per second.

If accuracy degradation in excess of 0.4 mil still exists, the barrel shall be replaced.

6.7 Standard deviation. The standard deviation for use in testing for projectile extraction, muzzle velocity, chamber pressure and action time shall be calculated in accordance with the following formula:

$$\sigma = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n-1}}$$

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6.8 Minimum cartridge weight. The minimum cartridge weight shall be determined for each lot based on the minimum propellant charge weight selected per 4.5.12. The formula for minimum cartridge weight, when assembled in accordance with Drawing 11825954 is defined below:

Minimum weight (grams) - 292 grams + charge weight (grams)

6.9 Muzzle velocity correction sample. A velocity of 800.2 meters per second is recorded in a test instrumentation configuration with the mid-point between the velocity coils located 30 meters from the muzzle. The muzzle velocity is calculated as follows:

Muzzle Velocity - 800.2 + (0.75 x 30) = 823 meters per second

6.10 Data reporting. Final data calculations shall be rounded off as follows:

Velocity - Nearest meter per second

Pressure - Nearest 0.1 MPa

Action time - Nearest 0.01 ms

Accuracy - Nearest 0.01 milliradian

6.11 Gun stoppage criteria. A gun stoppage is defined as failure to complete a single round cycle.

6.12 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate or organic pre-treatment shall be considered unsatisfactory if any of the following conditions exist.

a. Any area exceeding 60 square millimeters.

b. More than one area exceeding 30 square millimeters.

c. More than five areas of 15 square millimeters or less.

6.13 Rotating band failure criteria. Rotating band failure shall be defined as loss of any portion of the band which exceeds a 13MM length and has evidence of band seat serration imprint, or the wiping down to the bourellet diameter (no engraving in evidence) over the total length of the rotating band. For classification as a defect, there must be evidence, either by microflash photographs, recovery of the band or portions thereof, or by hole(s) in the witness board/screen. Normal band fringing is acceptable.

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6.14 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. The letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.15 Retest procedures. Failure of the material sample to comply with the velocity pressure and action time requirements shall result in a retest for the failed requirements. Sample size shall be as indicated in the applicable Table.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-C-63982 (AR) CARTRIDGE, 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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